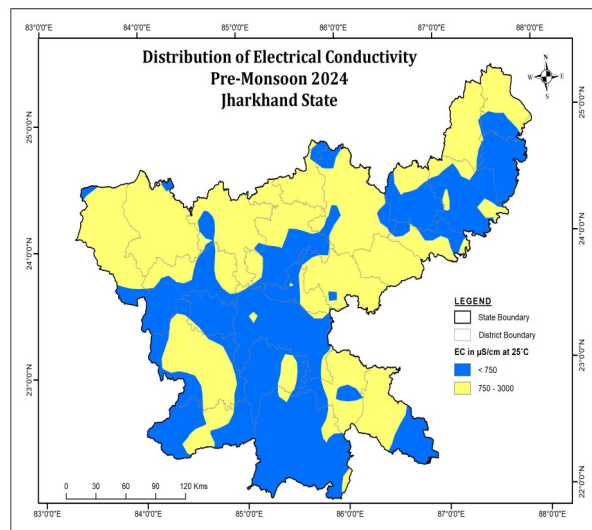


Groundwater Quality Scenario in Jharkhand

Parameters	No of samples	Permissible limit	No. of Samples above permissible limit	% Samples above permissible Limit
EC	381	3000 $\mu\text{S/cm}$	0	0.00
Fluoride	381	1.5 mg/L	15	3.94
Nitrate	381	45 mg/L	66	17.32



Districts with anomalous values at sporadic locations

EC (3000 $\mu\text{S/cm}$)	Not Any
Fluoride (F > 1.5 mg/L)	Bokaro, Chatra, Giridih, Godda, Hazaribagh, Koderma, Latehar, Palamu, Ramgarh
Nitrate (Nitrate > 45 mg/L)	Bokaro, Deoghar, Dhanbad, Dumka, Garhwa, Giridih, Godda, Gumla, Hazaribagh, Jamtara, Koderma, Latehar, Loherdaga, Palamu, Ranchi, Sahibganj, Simdega
Arsenic (As > 10 ppb)	Not Any

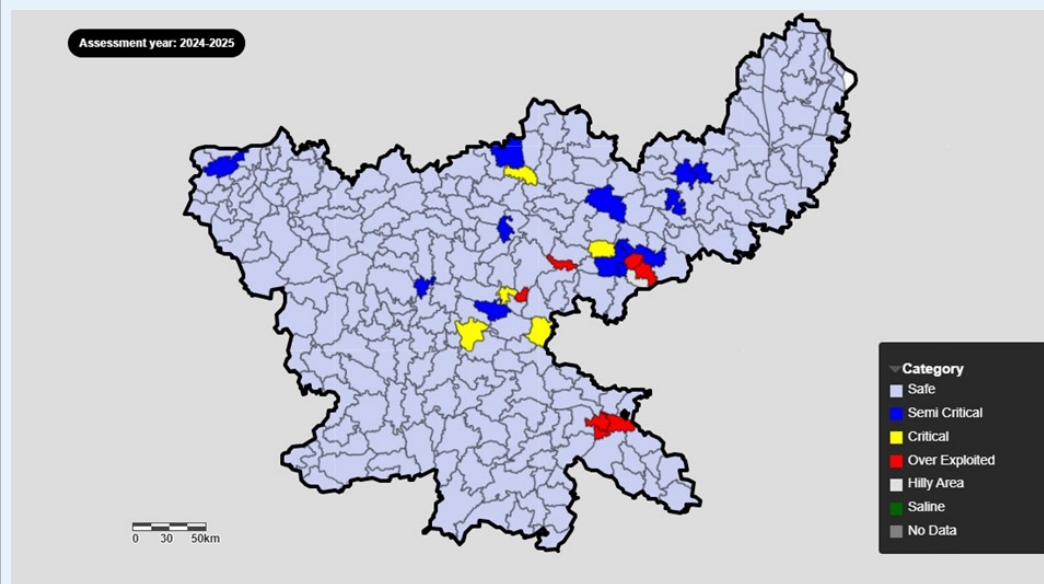
For Further Information, Contact to :
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NH IV Faridabad, Haryana - 121001
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Central Ground Water Board Department of Water Resources, RD & GR Ministry of Jal Shakti, Government of India



Dynamic Ground Water Resources, Ground Water Level & Ground Water Quality of Jharkhand, 2025

November, 2025

Groundwater Resource Scenario in Jharkhand

- Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Block).
- The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- GWRA- 2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad.

Salient Features

1	Rainfall	1,306.33 mm
2	Hydrogeology	Nearly 85 % of the State is underlain by hard rocks. Rest of the State is underlain by semi-consolidated formations and unconsolidated sediments.
3	Recharge Worthy Area of the State	60.65 Thousand Sq. Km
4	Assessment Unit (AU) Type / Number	Block / 263 Numbers
5	Average area of Assessment Unit	230.61 Sq. Km

Findings

	Attribute	GWRA-2017	GWRA-2020	GWRA-2022	GWRA-2023	GWRA-2024	GWRA-2025
1	Total Annual Ground Water Recharge (in bcm)	6.21	6.15	6.21	6.25	6.28	6.15
2	Annual Extractable Ground Water Resources (in bcm)	5.69	5.64	5.69	5.73	5.76	5.63
3	Annual Ground Water Extraction (in bcm)	1.58	1.64	1.78	1.8	1.81	1.85
4	Stage of Ground Water Extraction (in %)	27.73	29.13	31.35	31.38	31.43	32.89

bcm: Billion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

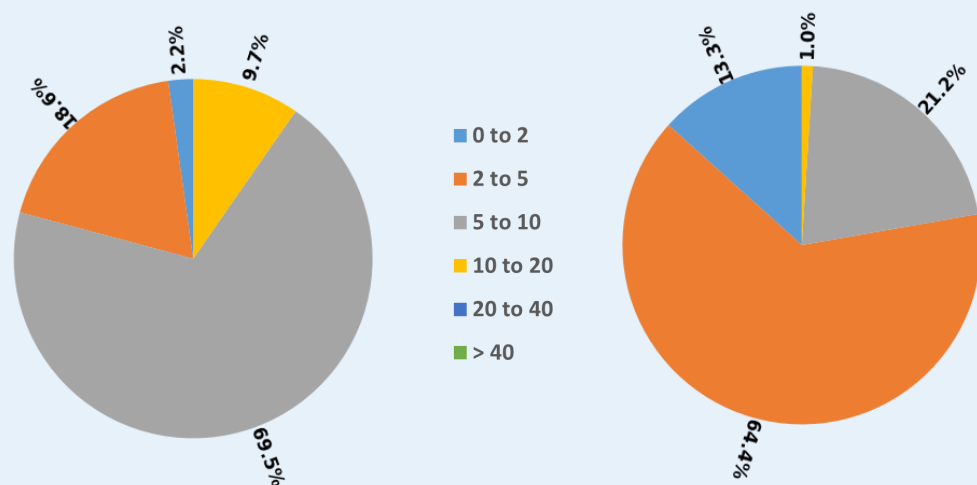
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs	Number of AUs	% of AUs
1	Safe	245	94	244	94	241	92	241	91.63	240	91.25	240	91.25
2	Semi-critical	10	4	10	4	11	4	11	4.18	12	4.56	12	4.56
3	Critical	2	1	2	1	6	2	6	2.28	6	2.28	5	1.90
4	Over-exploited	3	1	3	1	5	2	5	1.9	5	1.9	6	2.28
5	Saline												
Total number of AUs		260		259		263		263		263		263	

Recommendations:-

- The State is underlain by diverse rock types of different geological ages ranging from Archaean to Recent. The major rock types are igneous and metamorphic rocks covering nearly 85 percent of the geographical area of the state. The Total Annual Ground Water Recharge of the State has been assessed as 6.15 bcm and Annual Extractable Ground Water Resources is 5.63 bcm. And Stage of Extraction is 32.89 %. Out of 263 assessment units 6 units categorized as 'Over-exploited', 5 units (1.90 %) as 'Critical', 12 units (4.56 %) as 'Semi-critical' and rest 240 units (91.25%) are under 'Safe' category.
- More numbers of Water Harvesting and Conservation Structures may be constructed to catch the rain as the State is blessed with more than 1300 mm annual rainfall particularly in the hard rock terrain. State may also effectively use "Master plan for Artificial Recharge" prepared by CGWB in consultation with State Government (<https://cgwb.gov.in/cgwbpnmi/publication-detail/324>).
- Development of springs and their catchment in hilly areas for their sustainability.
- Restoration/rejuvenation of all the existing tanks should be taken up with the view of accommodating the available surface run off and thus augmentation of the ground water resources by artificial recharge. Periodical maintenance of these tanks is to be ensured.
- The "Manual on Artificial Recharge Techniques for augmentation of ground water" prepared by CGWB may be used for planning (<https://cgwb.gov.in/sites/default/files/MainLinks/Manual-Artificial-Recharge.pdf>).
- National Aquifer Mapping & Management Programme (NAQUIM) Reports prepared by CGWB (<https://cgwb.gov.in/cgwbpnmi/>) which are also being shared with State/District Authorities and Ground Water Year Book published by CGWB having water level & water quality data may be used in Ground water management (<https://cgwb.gov.in/cgwbpnmi/>). In the safe category areas of Jharkhand, State Government can judiciously develop the ground water resource mainly for agricultural use, however, at no point of time the extraction level should exceed 70%.
- Creating awareness (Mass Awareness Campaign for public and farmers, slideshows, display boards on water conservation, Water Management Training Programme for personnel related with water sector, painting/essay competition for school students etc.) regarding water conservation etc may be organized at appropriate level.
- State may review their free/subsidized electricity policy to farmers (if applicable), bring suitable water pricing policy and may work further towards crop rotation/diversification/other initiatives to reduce overdependence on groundwater.
- Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in Jharkhand

1. Depth to Groundwater Level

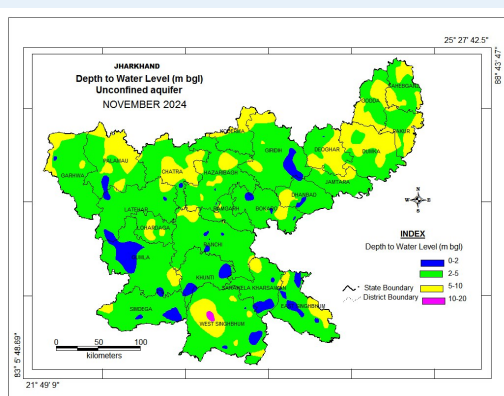
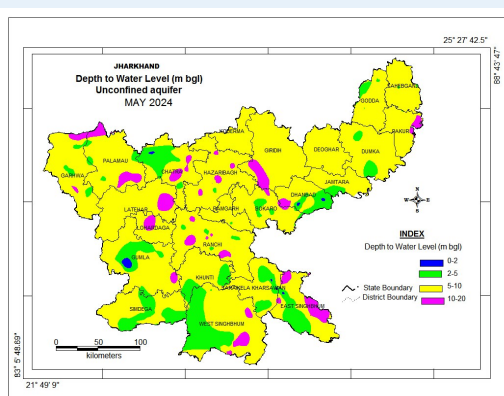


Pre-Monsoon (May 2024)

About 67% wells have depths of 5–10 m bgl, shallow levels (< 2 m) in 2 %, and deeper levels (10–20 m) in 14% of wells, indicating generally moderate groundwater conditions across the state.

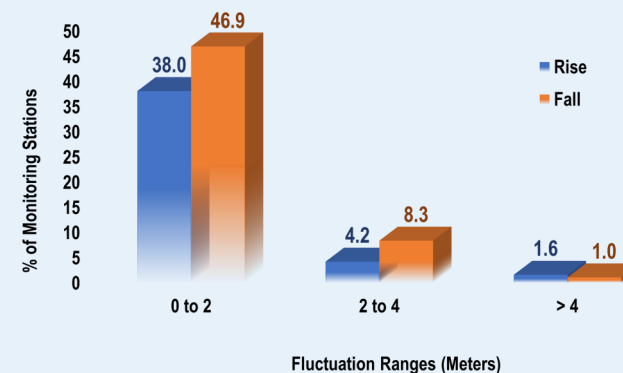
Post-Monsoon (November 2024)

Around 64% of wells fall in the 2–5 m range, about 14% show very shallow levels (< 2 m), and only ~ 1 % record depths beyond 10 m, reflecting largely stable and favorable groundwater conditions overall.



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

